

MAM2012S

Differential Equations (2DE)

University of Cape Town

July 27, 2026

1 Course information

Instructors: I am Mr Thomas van Heerden, lecturer for MAM2012S (2DE). My email address is `thomas.vanheerden@uct.ac.za`.

You're welcome to email me with queries but please make sure that you put "MAM2012S" in the subject line. I teach two courses in the second semester so I get a lot of student queries. If you don't put "MAM2012S" in the subject line and just ask "When is the first test?" I can't answer because I don't know which course you're talking about.

If the answer to your question is in the course information sheet or on AmaThuba I will not respond, otherwise I will try to get back to you within one working day.

Teaching: Teaching is face-to-face. Lectures are on Tuesday, Friday and some Wednesdays in M320 during 4th period.

We have MAM2012S lectures on the following Wednesdays: Aug 5, 26; Sep 16; Oct 7, 21.

Tutorials: Tutorials take place from 14:00 – 14:45 and 15:00 – 15:45 on Thursdays and Fridays. There are no face-to-face tutorials in the first week of the semester but there is a tutorial sheet that you should complete in your own time.

Tutorials are compulsory; you must attend a tutorial every week.

Please sign up for a tutorial slot on AmaThuba using the Groups tool.

Textbook: There is no prescribed book for this course. I've written up a set of notes that cover everything we're going to do in the course. You can download it from AmaThuba (under the "Content" tab, in the "General Information and Resources" folder). If you want extra material I like Zill's *A First Course in Differential Equations* for the material on ODEs, and Haberman's *Applied Partial Differential Equations* for the material on PDEs, but most of the introductory textbooks in the library will cover the same work. Nothing we're going to do is particularly unusual.

I will upload some of the problems (and solutions) from Zill to AmaThuba.

Syllabus: Linear ODEs, systems of linear ODEs and linear PDEs. See the table of contents in the notes for more detail.

Assessment: There are two class tests during the semester and an exam at the end of the year. In addition there will be tests during the tutorials. The first class test is on Monday 31 August at 18:00. The second is on Thursday 15 October at 18:00.

$$\text{Class Record} = 10\% \text{ Tutorial Tests} + 45\% \text{ Test 1 Score} + 45\% \text{ Test 2 Score}$$

and your final mark is the better of (20% Class record + 80% Exam) or (40% Class Record + 60% Exam).

You need a Class Record of at least 35% to be awarded your DP and allowed to write the exam. In addition, you need to attend all your tutorial sessions.

If you miss a test for good reason, please email me with an explanation (illness, death in the family, mugging, etc.) along with some evidence (medical certificate, death certificate, case number, etc.) as soon as possible. I appreciate that this may not be the day after the test if you are seriously ill, and that's okay.

Please note that departmental policy is to accept medical certificates that state “According to my knowledge” and not to accept certificates that state “As I was informed”. Please ask your practitioner to be clear on the certificate.

If you miss a test I may require you to be reassessed. Alternatively the remaining assessments may be reweighted to produce your Final Module Mark. If you miss all the assessments for a module you will have to sit an alternative assessment before the exam. This is normally an oral examination and I will liaise with you to arrange a mutually suitable time.

If you write a test the mark that you achieve is final. I cannot retroactively excuse you from the test. If there is a good reason why the test isn’t going to be an accurate reflection of your ability (e.g. illness, death in the family etc.) then don’t write it and send me the email with an explanation of your situation. If you do write the test I will treat the mark as a fair indicator of your progress.